

FIGURE 4 | PRODUCT-CONSENSUS GUARDRAIL LADDER

Satellite products and local records tier claims before policy interpretation

A Mascon and local-evidence ladder

regional_card	CSR	GSFC	GFZ raw	GFZ leakage	JPL CRI	Local	Claim class
NCP / Beijing	Sign+	Sign	Sign	Sign	Auth boundary	High mechanism	Mechanism anchor
Tokyo Bay / Yokohama	Material	Sign	Sign	Sign	Auth boundary	Rising wells	Sign-supported coastal case
Mumbai / Bhayandar	Material	Weak sign	Sign	Sign	Auth boundary	Contradicts recovery	Candidate-only boundary
Delhi	Material	Material	Sign	Sign	Auth boundary	Depletion	Product-material depletion
Lahore / Punjab	Material	Near/sign	Sign	Sign	Auth boundary	Borehole + InSAR	Regional depletion review

B Interpretation framework



SUPPORTED Coordinate utility

Products and local records define a review class.



BOUNDED Magnitude materiality

S_y and local aquifer evidence condition the threshold.



NOT SUPPORTED City-scale prediction

GRACE products do not become wells or engineering design data.

Raw agreement is not enough. The ladder asks: primary screen, independent product sign, leakage stress, authentication boundary, local hydrogeology, then claim class.